

β_2 Integrins differentially regulate $\gamma\delta$ T cell subset thymic development and peripheral maintenance

Claire L. McIntyre^a, Leticia Monin^b, Jesse C. Rop^a, Thomas D. Otto^a, Carl S. Goodyear^a, Adrian C. Hayday^{b,c}, and Vicky L. Morrison^{a,1}

^aInstitute of Infection, Immunity & Inflammation, University of Glasgow, G12 8TA Glasgow, United Kingdom; ^bImmunosurveillance Laboratory, The Francis Crick Institute, NW1 1AT London, United Kingdom; and ^cPeter Gorer Department of Immunobiology, King's College London, WC2R 2LS London, United Kingdom

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The $\gamma\delta$ T cells reside predominantly at barrier sites and play essential roles in immune protection against infection and cancer. Despite recent advances in the development of $\gamma\delta$ T cell immunotherapy, our understanding of the basic biology of these cells, including how their numbers are regulated in vivo, remains poor. This is particularly true for tissue-resident $\gamma\delta$ T cells. We have identified the β_2 family of integrins as regulators of $\gamma\delta$ T cells. β_2 -integrin-deficient mice displayed a striking increase in numbers of IL-17-producing V γ 6V δ 1⁺ $\gamma\delta$ T cells in the lungs, uterus, and circulation. Thymic development of this population was normal. However, single-cell RNA sequencing revealed the enrichment of genes associated with T cell survival and proliferation specifically in β_2 -integrin-deficient IL-17⁺ cells compared to their wild-type counterparts. Indeed, β_2 -integrin-deficient V γ 6⁺ cells from the lungs showed reduced apoptosis ex vivo, suggesting that increased survival contributes to the accumulation of these cells in β_2 -integrin-deficient tissues. Furthermore, our data revealed an unexpected role for β_2 integrins in promoting the thymic development of the IFN γ -producing CD27⁺ V γ 4⁺ $\gamma\delta$ T cell subset. Together, our data reveal that β_2 integrins are important regulators of $\gamma\delta$ T cell homeostasis, inhibiting the survival of IL-17-producing V γ 6V δ 1⁺ cells and promoting the thymic development of the IFN γ -producing V γ 4⁺ subset. Our study introduces unprecedented mechanisms of control for $\gamma\delta$ T cell subsets.

β_2 integrins | $\gamma\delta$ T cells | immune homeostasis

The $\gamma\delta$ subset of T cells are innate-like cells that play essential roles in immune surveillance, neonatal immunity, and responses to infection and tumors. Tissue-resident $\gamma\delta$ T cells localize within barrier sites such as the skin, gut, lungs, genital tract, and oral mucosa, enabling rapid responses against infectious challenge without the need for clonal expansion. Upon recognition of foreign antigens, host stress-induced markers, or innate-derived cytokines (1–3), $\gamma\delta$ T cells mediate immune protection via direct killing of target cells and production of the proinflammatory cytokines IFN γ and IL-17.

$\gamma\delta$ T cells are the major innate source of IL-17 during the early phase of an immune response (4). Mouse models have demonstrated protective roles for IL-17-producing $\gamma\delta$ T cells in a range of mucosal infections (5–7). In humans, IL-17-producing $\gamma\delta$ T cells have been reported in cord blood (8) and can be expanded from peripheral blood mononuclear cells in vitro (9, 10). Although their contribution to infection-driven immune responses remains poorly described, IL-17-producing $\gamma\delta$ T cell numbers are increased in patients with bacterial meningitis (10), suggesting an important role.

We set out to explore the regulatory roles of β_2 integrins in $\gamma\delta$ T cell populations that reside in barrier tissues. The β_2 integrins are leukocyte-specific adhesion molecules that consist of the β_2 chain (CD18) coupled to CD11a (α_L), CD11b (α_M), CD11c (α_X), or CD11d (α_D). It is well known that Leukocyte Function-associated Antigen-1 (LFA-1; CD11a/CD18) is essential for many aspects of $\alpha\beta$ T cell function (11): by binding intercellular adhesion molecules on endothelial cells, LFA-1 mediates T cell

recruitment to lymph nodes (LN) and sites of inflammation; LFA-1 localizes to the immunological synapse between interacting T and antigen-presenting cells, stabilizing contacts and lowering the threshold for T cell activation; and T cell LFA-1 mediates interactions with target cells (e.g., B cells or cancerous/infected cells), facilitating effector T cell functions. However, the involvement of β_2 integrins in $\gamma\delta$ T cell recruitment, interactions, and function is less well defined (reviewed in ref. 12). While some reports demonstrate a role for β_2 integrins in mediating $\gamma\delta$ T cell interactions with endothelial cells (13), $\gamma\delta$ T cell trafficking to sites of inflammation can occur independently of β_2 integrins, as shown in multiple sclerosis (14) and psoriasis (15) models.

Evidence suggests that β_2 integrins control $\gamma\delta$ T cell numbers and/or localization: under conditions of skin and periodontal inflammation in CD18-deficient (CD18^{hypos} PL/J) and CD11a knockout (KO) mice, respectively, increased numbers of $\gamma\delta$ T cells were found at inflamed sites (15, 16). Even in the absence of overt inflammation, increased proportions and/or numbers of $\gamma\delta$ T cells have been reported in the cervical LNs (17), spleen, and mesenteric LNs (18) of CD18 KO mice. Based on this evidence, we hypothesized that β_2 integrins regulate $\gamma\delta$ T cell numbers during homeostasis and inflammation. However, these studies have been limited in the tissues analyzed, making it unclear if the alterations in cell number are due to expansion or redistribution. Additionally, $\gamma\delta$ T cell numbers, subset profiles, and phenotype at barrier sites (e.g., lungs, small intestine, uterus,

Significance

$\gamma\delta$ T cells reside in barrier tissues and provide immune protection against infection and cancer. Their antitumor potential has led to recent advances in the development of $\gamma\delta$ T cell immunotherapy. However, our understanding of the basic biology of these cells, including which molecules and pathways control their maintenance within barrier tissues, remains poor. We demonstrate that β_2 integrin adhesion molecules play a major role in regulating $\gamma\delta$ T cell subset numbers during homeostasis: the loss of β_2 integrin expression results in a striking increase in IL-17-producing $\gamma\delta$ T cells in the lungs and uterus due to enhanced survival. These findings illustrate a mechanism of $\gamma\delta$ T cell regulation that may have significant implications for immunotherapy development.

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The authors declare no competing interest.

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¹To whom correspondence may be addressed. Email: vicky.morrison@glasgow.ac.uk.

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gingiva), where these cells have major immune protective roles, have not been analyzed in β_2 integrin-deficient mice.

In this study we investigated the impact of β_2 integrin deficiency on $\gamma\delta$ T cell subset numbers, localization, and phenotype. We demonstrate that loss of β_2 integrin expression leads to a tissue-restricted increase in IL-17-producing V γ 6V δ 1⁺ $\gamma\delta$ T cells that is prominent in the lungs, uterus, and spleen. We also reveal a role for β_2 integrins in promoting thymic development of IFN γ -producing CD27⁺ V γ 4⁺ $\gamma\delta$ T cells.

Results

β_2 Integrin-Deficient Mice Have a Tissue-Restricted Increase in $\gamma\delta$ T Cells. $\gamma\delta$ T cell numbers were quantified in a range of lymphoid and mucosal tissues from β_2 -integrin-deficient (CD18 KO) mice. $\alpha\beta$ T cells were analyzed for comparison. $\gamma\delta$ T cell numbers were increased in the lungs, uterus, spleen, and blood of CD18 KO mice, whereas numbers in the skin (including both the dermis and epidermis), oral mucosa, gut, and other lymphoid tissues were unaffected by β_2 integrin loss (Fig. 1; gating shown in *SI Appendix, Fig. S1*). As anticipated, $\alpha\beta$ T cell numbers were reduced in the LNs, Peyer's patches, and bone marrow as a result of defective homing (19, 20). Importantly, we saw no reduction in $\gamma\delta$ T cell numbers in any of the tissues analyzed, indicating that the increase in $\gamma\delta$ T cells in CD18 KO mice is not due to redistribution but, instead, to a tissue-specific expansion.

β_2 Integrin Loss Leads to an Increase in IL-17-Producing $\gamma\delta$ T Cells. $\gamma\delta$ T cells are broadly divided into two mutually exclusive phenotypes based on a cytokine profile of IFN γ -producing cells and IL-17-producing cells. Splenic $\gamma\delta$ T cells from CD18 KO mice produced significantly higher levels of IL-17 as both proportion and absolute numbers compared to WT cells (Fig. 2*A* and *B*). Increased IL-17 was observed across the tissues where $\gamma\delta$ T cell numbers were increased (Fig. 2*B*). The enhanced numbers of IL-17-producing $\gamma\delta$ T cells in CD18 KO mice contributed to an increase in serum IL-17 levels (*SI Appendix, Fig. S2 A and B*). Despite the apparent defect in the percentage of IFN γ -producing CD18 KO $\gamma\delta$ T cells (Fig. 2*A*), the number of IFN γ -producing $\gamma\delta$ T cells was equivalent (Fig. 2*B*).

The Expanded $\gamma\delta$ T Cell Population in β_2 -Integrin-Deficient Mice Expresses the V γ 6V δ 1 TCR. Next, we questioned whether the expanded $\gamma\delta$ T cell population was due to an increase in one or more $\gamma\delta$ T cell subsets in β_2 -integrin-deficient mice. Murine $\gamma\delta$ T cell subsets express different V γ chains, numbered V γ 1 to 7 [Heilig and Tonegawa nomenclature (21)], which localize to specific tissues and have different capacity for cytokine production (22). We predicted that CD18 KO mice would have an increase in V γ 4⁺ and/or V γ 6⁺ $\gamma\delta$ T cells, as these subsets are the predominant IL-17 producers (23). V γ chain expression was initially analyzed using antibodies against V γ 1, V γ 4, and V γ 5. Despite proportional differences in multiple subsets (*SI Appendix, Fig. S3A*), calculation of absolute numbers revealed a significant increase only in the V γ 1⁺V γ 4⁺V γ 5⁺ $\gamma\delta$ T cells in CD18 KO mice (Fig. 2*C*). This was true for all tissues where an increase was previously found in total $\gamma\delta$ T cell numbers (Fig. 1 and Fig. 2*C*). PCR analysis of the "unknown" V γ 1⁺V γ 4⁺ population in CD18 KO mice revealed expression of the V γ 6 and V δ 1 TCR chains (Fig. 2*D* and *SI Appendix, Fig. S3B*). Note that, although V γ 2 is expressed at the messenger RNA level, protein expression does not correlate with in-frame transcript expression, and a high RNA level may reflect poor allelic exclusion (24). Extensive flow cytometric analysis confirmed that CD18 KO splenic $\gamma\delta$ T cells displayed a phenotype consistent with IL-17-producing cells: they were predominantly CD62L^{lo}, CD44^{hi}, CD27⁺, CD45RB⁺, and ROR γ t⁺ (*SI Appendix, Fig. S3C*), suggestive of IL-17 production (25, 26), and they expressed higher levels of CD3 and TCR $\gamma\delta$ than their WT counterparts (*SI Appendix, Fig. S3C*), a

phenotype that implies a V γ 6-expressing phenotype (27). Together, these data show that the absence of β_2 integrin expression leads to an increase in IL-17-producing $\gamma\delta$ T cells that express the V γ 6V δ 1 TCR.

The Increase in IL-17-Producing $\gamma\delta$ T Cells Is due to Loss of LFA-1. β_2 Integrins are a family of four receptors, the loss of one or more of which could cause the $\gamma\delta$ T cell expansion seen in CD18 KO mice. Analysis of CD11 chain KO mice revealed that the increase in $\gamma\delta$ T cell numbers was due to the specific loss of LFA-1, as increased numbers of $\gamma\delta$ T cells were detected in the spleen and lungs of CD11a KO mice but not in mice lacking CD11b, CD11c, or CD11d (*SI Appendix, Fig. S4A*). We confirmed that the enhanced capacity to produce IL-17 was also observed in the expanded $\gamma\delta$ T cell population in CD11a KO mice (*SI Appendix, Fig. S4B*).

Thymic Development of V γ 6⁺ $\gamma\delta$ T Cells Is Unaffected by β_2 Integrin Loss, while CD27⁺V γ 4⁺ $\gamma\delta$ T Cell Development Is Impaired. We next sought to investigate the mechanism of expansion of V γ 6⁺ IL-17-producing $\gamma\delta$ T cells in β_2 -integrin-deficient mice. V γ 6⁺ $\gamma\delta$ T cells egress from the thymus from embryonic day 16.5 (E16.5) to a few days after birth and, thereafter, are maintained in the periphery by self-renewal (28). Analysis of the thymus showed equivalent numbers of V γ 6⁺ $\gamma\delta$ T cells in WT and CD18 KO mice in embryonic, neonatal (1- and 3-d-old), and adult (4-wk-old) mice (Fig. 3*A*). As thymic output of V γ 6⁺ cells both at the peak of development and in adult mice was normal, we conclude that the development of V γ 6⁺ cells is not influenced by β_2 integrin deficiency.

Surprisingly, while V γ 1⁺ and V γ 5⁺ $\gamma\delta$ T cell development was also equivalent in WT and CD18 KO mice, we found a reduction in the number of thymic V γ 4⁺ $\gamma\delta$ T cells in embryonic and neonatal mice (Fig. 3*A*). Further analysis to differentiate between immature and mature populations revealed that the reduced number of total V γ 4⁺ cells was due to a loss of immature CD24⁺ cells at E18.5 and to a loss of mature CD24⁺ V γ 4⁺ cells in 4-wk-old mice (Fig. 3*B*). This suggests that the loss of β_2 integrin expression impairs V γ 4⁺ $\gamma\delta$ T cell development from the early immature stage. The number of V γ 6⁺ cells, regardless of maturation status, was similar in the thymus of WT and CD18 KO mice, further confirming that V γ 6⁺ $\gamma\delta$ T cell development is unaffected by β_2 integrin loss.

IL-17- and IFN γ -producing $\gamma\delta$ T cells can be distinguished by expression of CD27, which is present only on IFN γ producers (25). Further analysis revealed that numbers of CD27⁺ V γ 4⁺ cells in the thymus were reduced in CD18 KO mice across all ages of mice analyzed (Fig. 3*C*). No differences were found in CD27⁺ V γ 4⁺ cells. These data suggest that β_2 integrin loss causes a specific defect in the development of CD27⁺ V γ 4⁺ $\gamma\delta$ T cells that have the potential to produce IFN γ .

To determine whether this impaired development impacts on the numbers of V γ 4⁺ cells in the periphery, tissues were analyzed in adult mice. We previously showed that the number of V γ 4⁺ cells was equivalent in CD18 KO mice at sites where the expanded V γ 6⁺ cells were found (lungs, uterus, spleen, and blood) (Fig. 2*C*). However, as V γ 4⁺ cells commonly reside within lymphoid tissues, we performed a more thorough examination. The number of V γ 4⁺ cells significantly decreased in the peripheral LNs (inguinal, brachial, and axillary pooled). A similar trend of reduced V γ 4⁺ cell numbers was observed in the mediastinal LN and bone marrow, but this did not reach statistical significance (Fig. 3*D*). These findings indicate that the impaired development of V γ 4⁺ $\gamma\delta$ T cells in CD18 KO mice results in reduced numbers of these cells specifically in the peripheral LNs.

Single-Cell Sequencing Reveals Evidence of Enhanced Survival of β_2 -Integrin-Deficient V γ 6⁺ Cells. Next, single-cell RNA sequencing (scRNAseq) was used to investigate the possible mechanism(s) by which β_2 integrins control $\gamma\delta$ T cell numbers in the periphery.

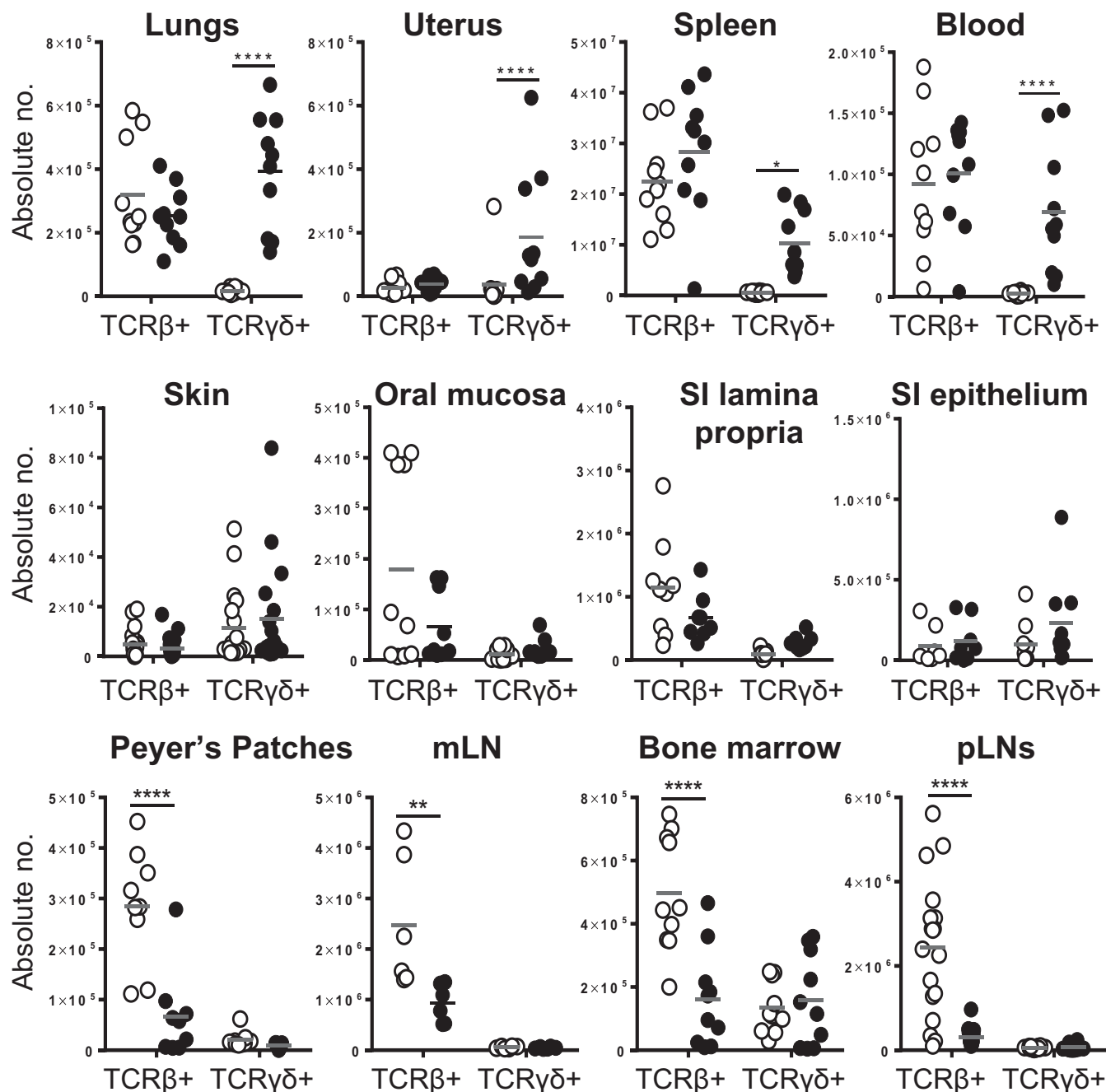


Fig. 1. CD18 KO mice have increased numbers of $\gamma\delta$ T cells in the lungs, uterus, spleen, and blood. Cells were isolated from various lymphoid and mucosal tissues of WT (open circles) and CD18 KO (closed circles) mice and then analyzed by flow cytometry. Skin includes both dermis and epidermis. Cells were gated on CD3⁺ and then on TCRβ⁺ or TCRγδ⁺ as shown in [SI Appendix, Fig. S1](#). Absolute number of $\alpha\beta$ and $\gamma\delta$ T cells per tissue is shown. For blood, numbers were calculated per 100 μ L. SI: small intestine; mLN: mesenteric LNs; pLN: peripheral LNs (brachial, axillary, and inguinal pooled). $n = 10$, data pooled from a minimum of three independent experiments; each symbol represents an individual mouse. Statistical differences were determined using a two-way ANOVA. * $P < 0.05$, ** $P < 0.01$, and **** $P < 0.0001$.

Principal component analysis of lung $\gamma\delta$ T cells revealed five biologically distinct clusters: the *Il17a*⁺ cluster which contains both V γ 6⁺ (*Cd163l1*⁺ *Trdv4*⁺) and V γ 4⁺ (*5830411N06Rik*⁺) cells, two *Cd27*⁺ clusters, namely *Ccr7*⁺ and *Ly6c2*⁺, and much smaller clusters of killer cells (*Gzma*⁺) and dividing cells (*Cdk6*⁺) (Fig. 4A and [SI Appendix, Fig. S5 A–C](#)). As expected, the CD18 KO cells were overrepresented in the *Il17a*⁺ cluster (Fig. 4B). High expression of the V γ 6 (*Tcrv-V6*) and V δ 1 (*Trdv4*) TCR subunits as well as Scart1 (*Cd163l1*) confirmed the expanded CD18 KO

population as V γ 6⁺ cells ([SI Appendix, Fig. S5D](#)). The number of differentially expressed (DE) genes when comparing WT to KO was much greater in the *Il17a*⁺ cluster (1,439 genes) than in the *Ccr7*⁺ (74 genes) or *Ly6c2*⁺ (32 genes) clusters (Fig. 4C and [Dataset S1 A–C](#)). We determined the DE genes that were unique to the IL-17-producing population (Fig. 4C and [Dataset S2](#)), excluding DE genes dependent on the loss of the β_2 integrin which were common to two or three of the dominant $\gamma\delta$ T cell populations. Pathway analysis revealed enrichment of genes associated

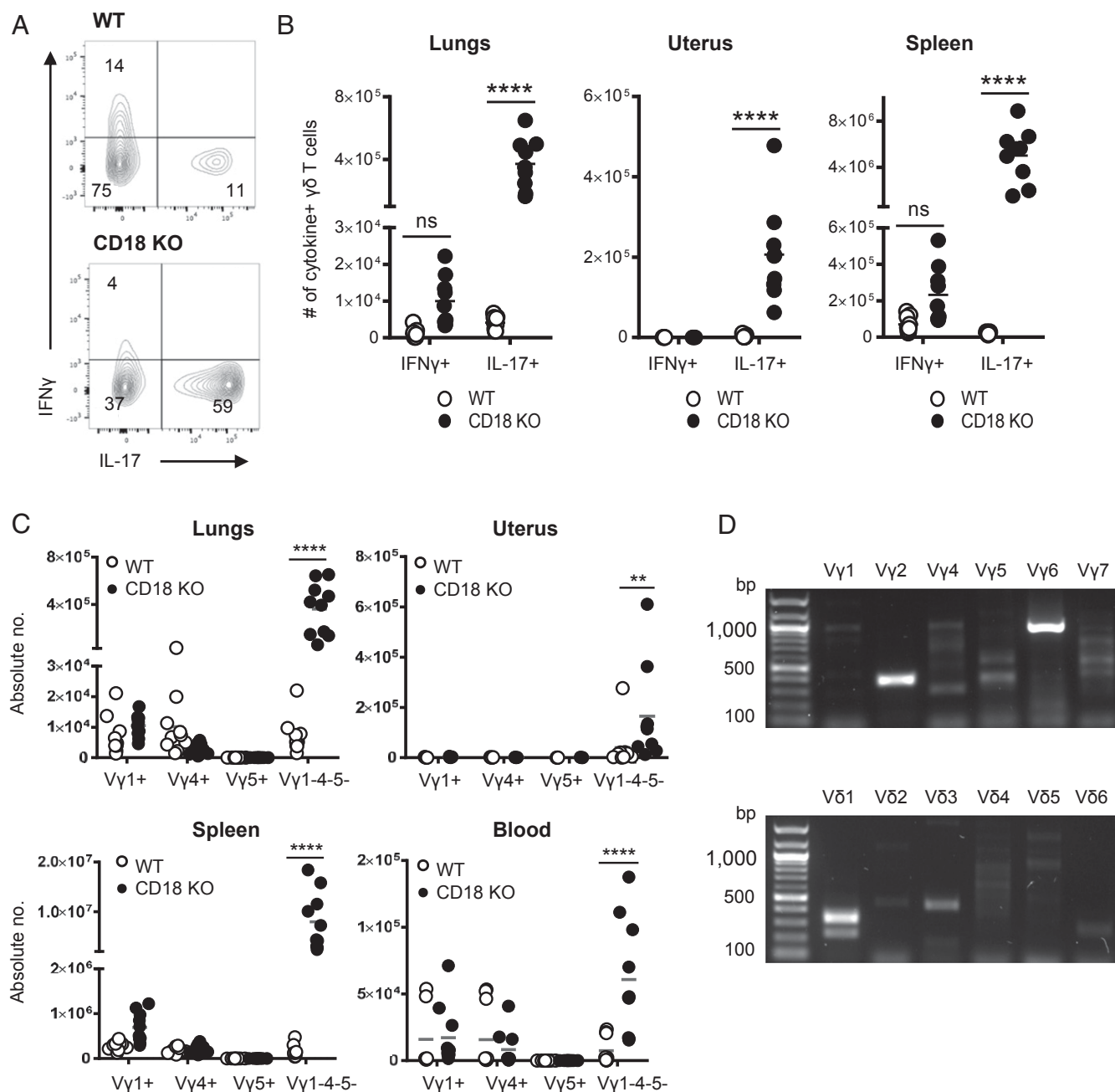


Fig. 2. IL-17-producing Vγ6Vδ1⁺ γδ T cells are increased in CD18 KO mice. (A) Representative plot showing percentage of IL-17⁺ and IFNγ⁺ γδ T cells from WT and CD18 KO spleen following in vitro stimulation with PMA/ionomycin/brefeldin A (BfA) for 4 h. (B) Number of IL-17⁺ and IFNγ⁺ γδ T cells from the lungs, uterus, and spleen of WT and CD18 KO mice following in vitro stimulation with PMA/ionomycin/BfA for 4 h. *n* = 9 pooled from three independent experiments. (C) Cells isolated from WT and CD18 KO tissues were analyzed for Vγ chain expression by flow cytometry. Samples were gated on CD3⁺TCRγδ⁺ and then on Vγ1⁺, Vγ4⁺, Vγ5⁺, or Vγ1⁺Vγ4⁺Vγ5⁺. Absolute number of each Vγ subset. *n* = 10 pooled from four independent experiments. (D) CD18 KO splenocytes were sorted by FACS for CD3⁺TCRγδ⁺Vγ1⁺Vγ4⁺ cells and then RNA was isolated and RT-PCR performed. Representative gels showing expression of Vγ and Vδ genes. Representative of *n* = 4. Positive controls for each Vγ and Vδ chain are shown in *SI Appendix, Fig. S3C*. Statistical differences were determined using a two-way ANOVA. ***P* < 0.01 and *****P* < 0.0001. Each symbol represents an individual mouse with the mean shown.

with T cell activation, survival, and proliferation in CD18 KO cells from the IL-17-producing cluster (Fig. 4D, *SI Appendix, Fig. S5E*, and *Dataset S3*). Specifically, the prosurvival genes *Bcl2a1a* and *Bcl2a1d*, known to support the survival of skin-resident Vγ6⁺ cells (29), were uniquely enriched in the CD18 KO *Il17a*⁺ γδ T cell cluster (Fig. 4E and *SI Appendix, Fig. S5E*). Other genes reported to enhance survival in T cells (TCR subunits and signaling molecules, the IL-2 receptor, *Cd44*, and *Icos*) as well as in cancer cells (*Crip1*)

(30) were also enriched in the CD18 KO *Il17a*⁺ cluster (Fig. 4E and *SI Appendix, Fig. S5E*). Together, scRNAseq analysis revealed the dysregulation of pathways involved in cell survival and proliferation specifically in the CD18 KO IL-17-producing γδ T cell population.

Cell Survival, rather than Proliferation, Is the Dominant Driver of Vγ6⁺ γδ T Cell Accumulation in β₂-Integrin-Deficient Mice. Assessing proliferation in vivo, we detected a small but significant increase

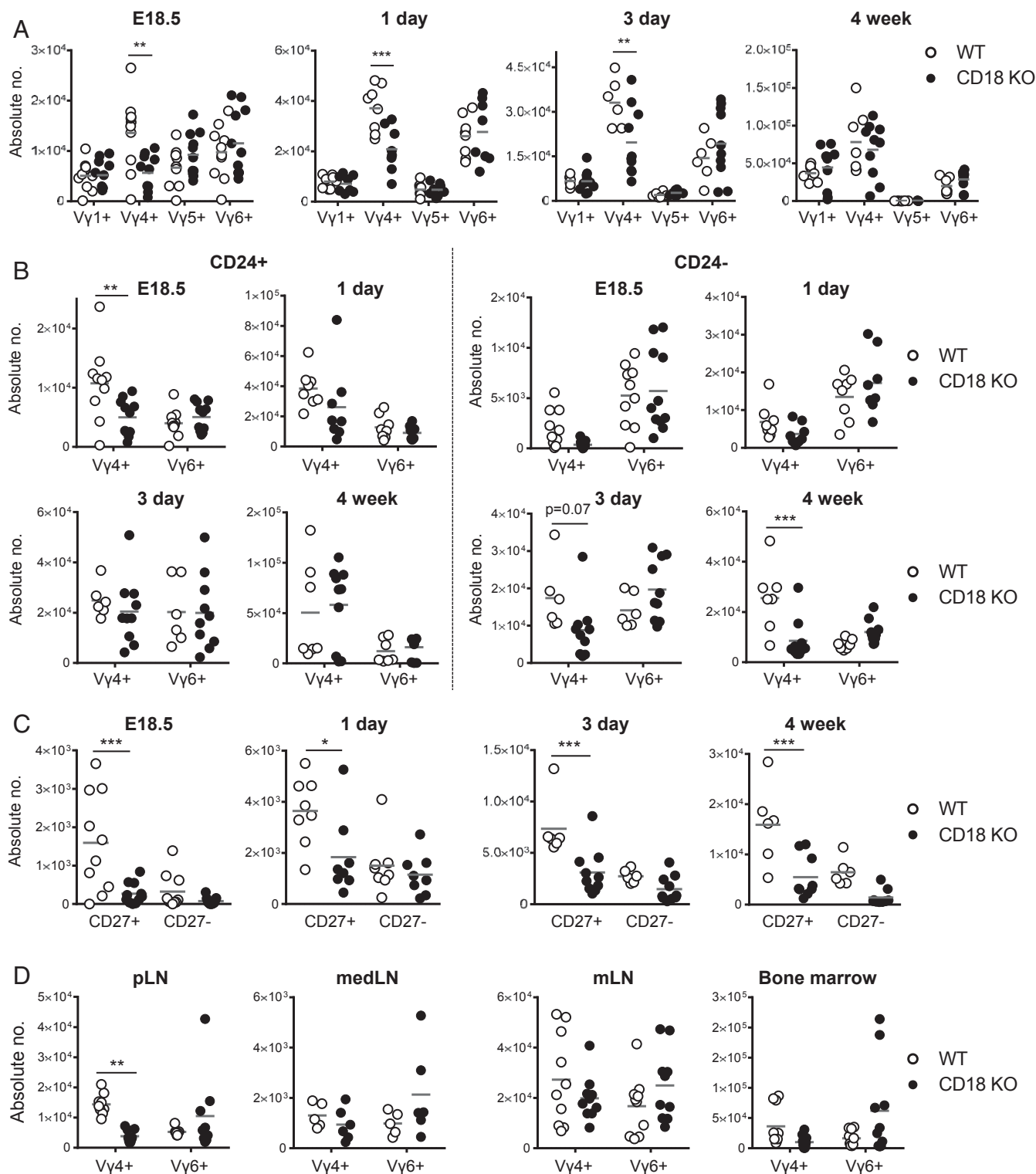


Fig. 3. CD18 KO mice show no difference in V γ 6 $^{+}$ thymic development but have impaired development of V γ 4 $^{+}$ $\gamma\delta$ T cells. (A–C) Thymocytes were analyzed from E18.5-, 1-d, 3-d, and 4-wk-old WT and CD18 KO mice by flow cytometry. Samples were gated on CD3 $^{+}$ TCR $\gamma\delta^{+}$ and then on V γ 1 $^{+}$, V γ 4 $^{+}$, V γ 5 $^{+}$, or V γ 1 $^{-}$ V γ 4 $^{-}$ V γ 5 $^{-}$ (V γ 6 $^{+}$). (A) Absolute number of each V γ subset. (B) Absolute number of V γ 4 $^{+}$ and V γ 6 $^{+}$ cells gated on either immature CD24 $^{+}$ (Left) or mature CD24 $^{-}$ (Right). (C) Absolute number of mature CD24 $^{-}$ V γ 4 $^{+}$ $\gamma\delta$ T cells gated on either CD27 $^{+}$ or CD27 $^{-}$. (D) Absolute number of V γ 4 $^{+}$ and V γ 6 $^{+}$ $\gamma\delta$ T cells from lymphoid tissues of adult mice. pLN: peripheral lymph nodes (inguinal, brachial, and axillary pooled); medLN: mediastinal lymph node; mLN: mesenteric lymph nodes. (A–C) $n = 6$ to 11 pooled from a minimum of three independent experiments. (D) pLN, mLN, and bone marrow: $n = 9$, data pooled from three independent experiments. medLN $n = 6$; data pooled from two independent experiments. Each symbol represents an individual mouse. Statistical differences were determined using a two-way ANOVA. * $P < 0.05$, ** $P < 0.01$, and *** $P < 0.001$.

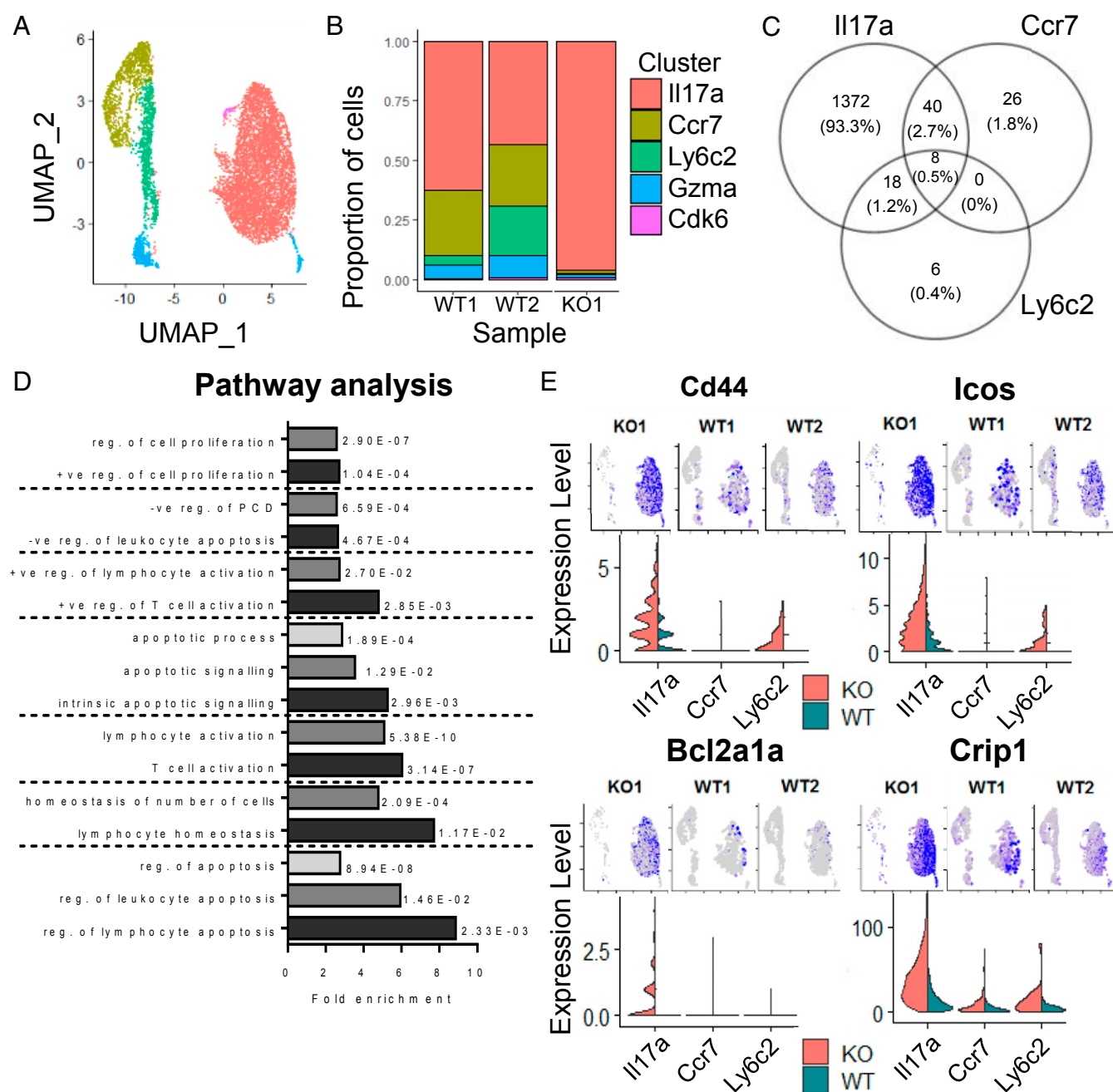


Fig. 4. Single-cell RNA sequencing reveals enhanced expression of genes known to negatively regulate apoptosis, specifically in $V\gamma 6^+$ cells from CD18 KO mice. $\gamma\delta$ T cells were purified from the lungs of two WT and one CD18 KO mouse by FACS. scRNAseq was performed and data analyzed as described in detail in *Materials and Methods*. The number of cells sequenced were the following: WT1, 918 cells; WT2, 6,315 cells; KO, 4,034 cells. (A) Principal component analysis plot showing the five identified clusters. (B) Frequency of cells in each of the identified clusters. (C) Venn diagram showing the shared and unique differentially expressed genes between WT and KO in the three major clusters, namely *II17a*⁺, *Ccr7*⁺, and *Ly6c2*⁺. (D) The apoptosis pathways identified in GO enrichment analysis of the differentially expressed genes unique to the *II17a*⁺ cluster. Fold enrichment and *P* value are shown for each pathway. +ve: positive; -ve: negative; reg.: regulation; PCD: programmed cell death. (E) Expression levels of selected genes identified by pathway analysis as associated with the negative regulation of leukocyte apoptosis.

in Ki67 expression in $V\gamma 6^+$ cells from both the spleen and lungs of CD18 KO mice compared to WT (Fig. 5A). However, we found similar incorporation of bromodeoxyuridine (BrdU) by WT or CD18 KO $V\gamma 6^+$ or $V\gamma 4^+$ cells in either tissue over the 7-d period of analysis (Fig. 5B), suggesting that β_2 -integrin-deficient $V\gamma 6^+$ cells do not show enhanced proliferation in adult mice. These data are supported by the scRNAseq analysis which shows that CD18 KO cells are not enriched in the dividing *Cdk6*⁺

cluster (Fig. 4B) and by gene ontology (GO) enrichment analysis where pathways associated with apoptosis dominated (Fig. 4D).

Next, we sought to determine whether cell survival is influenced by β_2 integrin deficiency. Detailed analysis of tissues from CD18 KO mice of various ages revealed that $V\gamma 6^+$ $\gamma\delta$ T cell numbers in the lungs and spleen were equivalent to WT mice at E18.5, when this population begins to egress the thymus (Fig. 6A and *SI Appendix, Fig. S6A*). However, at 1-d post birth, CD18

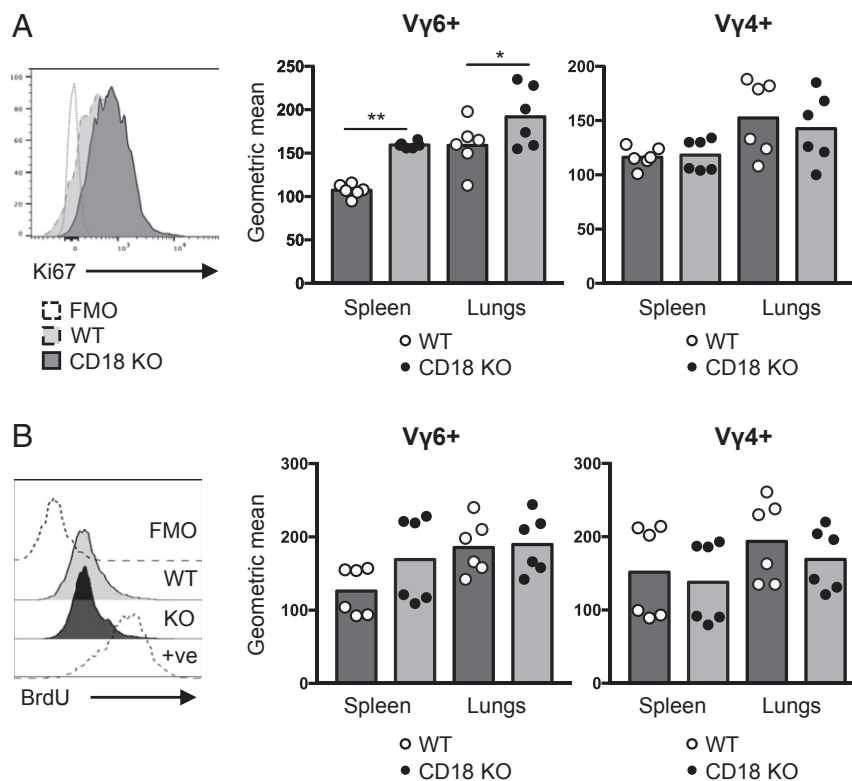


Fig. 5. No evidence of enhanced proliferation in Vγ6⁺ γδ T cells from CD18 KO mice. (A) Cells isolated from the spleen and lungs of WT and CD18 KO mice were stained for the expression of Ki67. Representative histograms are shown from the spleen. Pooled data show $n = 6$ from two independent experiments. (B) WT and CD18 KO mice were injected with 1 mg BrdU intraperitoneally and then fed 0.8 mg/mL BrdU in the drinking water for 7 d. Cells isolated from the spleen and lungs were analyzed by flow cytometry for expression of BrdU. Representative histograms are shown from the spleen. Pooled data show $n = 6$ from two independent experiments. +ve: positive control (thymus). Each symbol represents an individual mouse. Statistical differences were calculated using a two-way ANOVA. * $P < 0.05$ and ** $P < 0.01$.

KO mice showed increased numbers of Vγ6⁺ γδ T cells in the lungs, and from 3-d-old mice onward, the numbers in the spleen were also elevated. Interestingly, while Vγ6⁺ γδ T cell numbers in the lungs reached a plateau from day 3, numbers in the spleen of CD18 KO mice continue to rise as the mice age, increasing to 40-fold more than WT at 4 wk (Fig. 6A). Surface marker analysis confirmed that the expanded γδ T cell population displayed the expected CD45RB⁺ CD44⁺ CD27⁺ phenotype consistent with IL-17-producing potential (SI Appendix, Fig. S6B). These data indicate that the increased number of Vγ6⁺ γδ T cells initially arise in the lungs and then the spleen. The accumulation of cells over time is consistent with enhanced Vγ6⁺ cell survival in CD18 KO mice.

IL-7 is known to drive T cell survival through Bcl2 (32). As *Il7r* was expressed at higher levels in CD18 KO cells compared to WT in the *Il17a*⁺ cluster (SI Appendix, Fig. S5D) and “cellular response to IL-7” was a hit in pathway analysis (Dataset S3), we assessed expression of IL-7R and Bcl2 in lung γδ T cells. Expression of IL-7R was higher in Vγ6⁺ cells compared to Vγ4⁺ cells but was equivalent between WT and CD18 KO cells (SI Appendix, Fig. S7A). Bcl2 expression was also equivalent in WT and CD18 KO γδ T cells (SI Appendix, Fig. S7B). Thus, signaling through the IL-7R-Bcl2 axis is not elevated in β₂-integrin-deficient γδ T cells.

Next, we investigated the effect of β₂ integrin loss on γδ T cell survival by direct ex vivo analysis. Vγ6⁺ γδ T cells from the lungs of CD18 KO mice displayed reduced apoptosis compared to WT, with increased proportion of live cells and reduced proportion of late apoptotic cells (Fig. 6B). This effect was specific to the Vγ6⁺ cells, as the proportions of live and apoptotic cells in the Vγ4⁺ population were equivalent. Furthermore, the enhanced survival of CD18 KO cells was specific to the lungs, as

cells in the spleen showed similar levels of apoptosis regardless of genotype or subset (Fig. 6B). Of note, this enhanced survival was most apparent in 4-wk-old mice (Fig. 6C) as well as adult mice (Fig. 6B), whereas viability of Vγ6⁺ cells in neonatal mice was equivalent in WT and CD18 KO mice (Fig. 6C). Thus, the enhanced survival of lung Vγ6⁺ cells, particularly in adult mice, likely contributes to the accumulation of this γδ T cell subset in CD18 KO mice.

LFA-1 Expression Is Higher in Vγ6⁺ Cells Compared to Vγ4⁺ Cells, Particularly in the Lungs, in WT Mice. Finally, we questioned why β₂ integrin loss led to the selective expansion of the Vγ6⁺ subset. Assessment of β₂ integrin expression in γδ T cells revealed only CD11a (LFA-1) and not CD11b or CD11c expression (SI Appendix, Fig. S8). In the lungs, expression of CD11a was highest in the Vγ6⁺ population compared to Vγ1⁺ and Vγ4⁺ cells, whereas γδ T cell populations expressed equivalent, lower levels of CD11a in the spleen (Fig. 6C). CD11a expression was equivalent across γδ T cell populations in the peripheral LN (SI Appendix, Fig. S8) where Vγ6⁺ cell numbers are unaffected by CD18 deficiency (Fig. 3D). These data suggest that Vγ6⁺ γδ T cells in the lungs may be most affected by β₂ integrin loss as this population expresses the highest levels of LFA-1.

Discussion

In this study we reveal that the absence CD18 expression in mice leads to a striking increase in Vγ6Vδ1⁺ IL-17-producing γδ T cell numbers in the lungs and uterus, as well as in the blood and spleen. Vγ6⁺ cells showed no difference in thymic development or proliferation but displayed a transcriptional profile consistent with enhanced survival and reduced levels of apoptosis

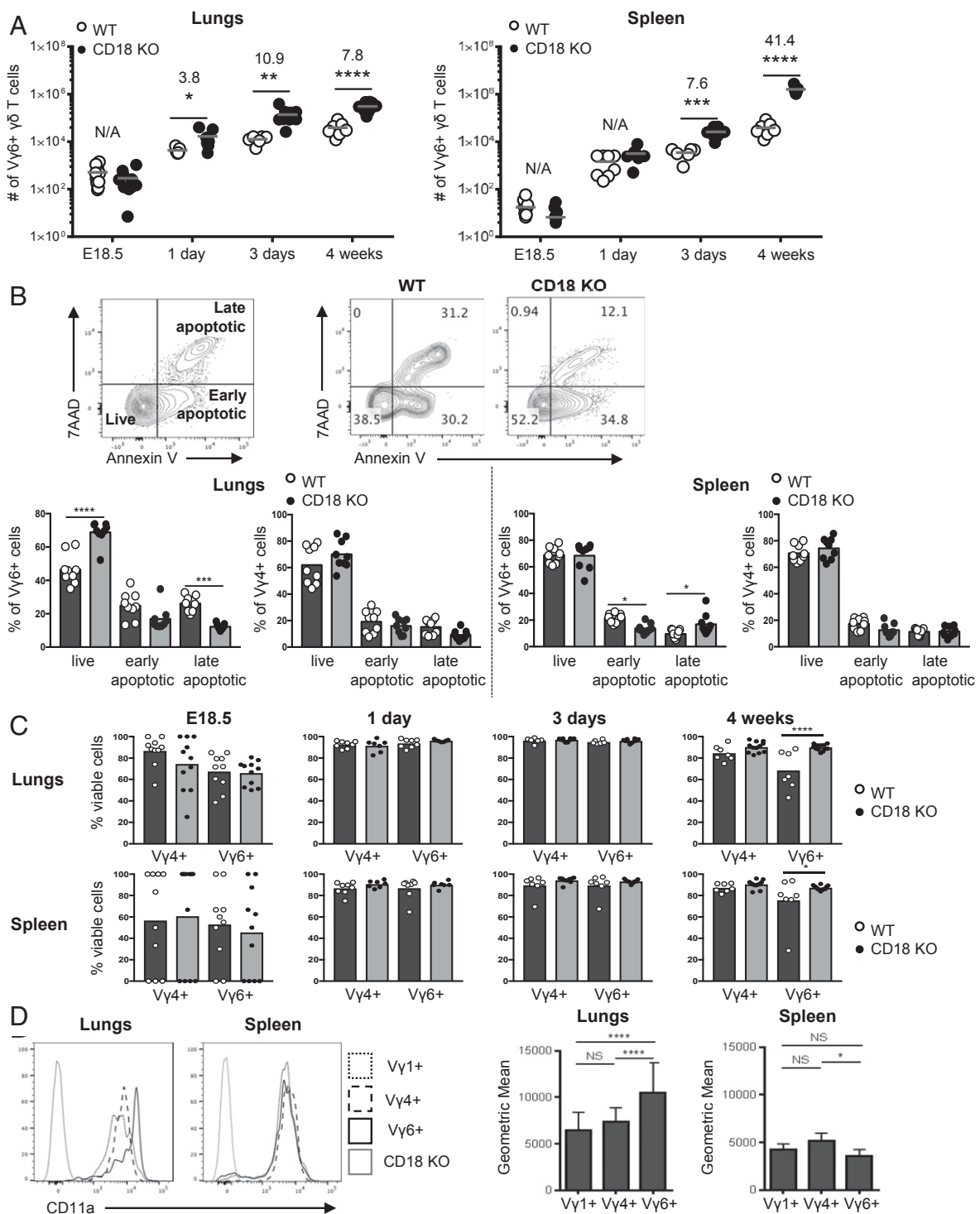


Fig. 6. CD18 KO Vγ6+ γδ T cells display enhanced survival. (A) Analysis of γδ T cells from the lungs and spleen of WT and CD18 KO mice at various ages. Cells were gated on CD3+ TCRγδ+ and then on Vγ1+, Vγ4+, Vγ5+, or Vγ1~Vγ4~Vγ5~ (Vγ6+). Absolute number of Vγ6+ (Vγ1~Vγ4~Vγ5~) γδ T cells is shown. Where significant differences between WT and CD18 KO were found, the fold difference is shown. WT spleen and lungs at 3 d old; $n = 6$. CD18 KO spleen at 1 d old, WT/CD18 KO lungs and WT/CD18 KO spleen at 4 wk old, WT lungs at 4 wk old; $n = 7$. WT spleen at 1 d old $n = 8$. CD18 KO spleen at 3 d old, CD18 KO lungs at 3 d/4 wk old, WT spleen at E18.5; $n = 10$. CD18 KO spleen and lungs at E18.5; $n = 11$. Data pooled from four to six litters with each litter run as an independent experiment; each symbol represents an individual mouse. Statistical differences were determined using a Student's t test with the false-discovery-rate approach two-stage linear step-up procedure of Benjamini, Krieger, and Yekutieli (31). (B) Cells isolated from adult WT and CD18 KO lungs and spleen were analyzed by flow cytometry for expression of apoptosis markers. Cells were gated on CD3+ TCRγδ+ and on then Vγ4+ or Vγ1~Vγ4~ (Vγ6+). Proportions of Vγ6+ or Vγ4+ γδ T cells that were live (Annexin V⁻ 7AAD⁻), early apoptotic (Annexin V⁺ 7AAD⁻), or late apoptotic (Annexin V⁺ 7AAD⁺) are shown. Graphs show $n = 9$ pooled from three independent experiments; each symbol represents an individual mouse. (C) Viable cells (staining negative for viability dye) were calculated in each γδ T cell subset at various ages of mice. n is as described in A. (D) CD11a expression in lung and spleen γδ T cell subsets was quantified by flow cytometry. $n = 8$; three independent experiments. Statistical differences were determined using a two-way ANOVA. NS: not significant, * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, and **** $P < 0.0001$.

ex vivo. Surprisingly, we also found that β_2 integrin loss impaired the development of IFN γ -producing (CD27 $^+$) V γ 4 $^+$ $\gamma\delta$ T cells, leading to reduced numbers of this subset in peripheral LNs. Together, our findings highlight roles for β_2 integrins in promoting V γ 4 $^+$ $\gamma\delta$ T cell development and in negatively regulating V γ 6 $^+$ $\gamma\delta$ T cell numbers in the periphery via apoptosis. In doing so, β_2 integrins alter the balance between IL-17- and IFN γ -producing $\gamma\delta$ T cells at mucosal and lymphoid sites.

Importantly, we found no defect in $\gamma\delta$ T cell numbers in any tissue, showing conclusively that the increase in $\gamma\delta$ T cell numbers reported here and in previous studies in integrin-deficient mice (17, 18) is not a redistribution effect but a tissue-specific increase in $\gamma\delta$ T cells that occurs in the absence of LFA-1. In agreement with previous studies (14, 15), our findings indicate that homing of $\gamma\delta$ T cells to an extensive range of mucosal and lymphoid tissues is not dependent on β_2 integrins.

It was not surprising that the increase in $\gamma\delta$ T cell numbers was most apparent in the lungs and uterus (in addition to the spleen), as V γ 6 $^+$ cells preferentially localize to these tissues (26, 33). However, V γ 6 $^+$ cells also reside in the dermis (34) and gingiva (35), where no differences were found. Analysis of whole skin (epidermis and dermis) may have masked any differences exclusive to the dermis. In addition, there may be stronger competition by other skin-resident T cells, precluding accumulation of V γ 6 $^+$ cells in the dermis. Alternatively, the expanded V γ 6 $^+$ $\gamma\delta$ T cell population in CD18 KO mice may be truly tissue specific. V γ 6 $^+$ cells are not described to preferentially home to the spleen, although they have been shown to reside there (26, 27). We propose that the increasing number of V γ 6 $^+$ cells exceeds the capacity within their resident tissues (lungs and uterus), which forces the $\gamma\delta$ T cells into the circulation and causes accumulation in the spleen. The enhanced number of $\gamma\delta$ T cells in the blood and appearance of increased numbers in the lungs prior to appearance in the spleen in newborn mice support this.

Analysis of CD18^{hy γ} mice (CD18 expression at 2 to 16% of WT) revealed the expanded $\gamma\delta$ population in the skin to have a CD27 $^-$ IL-17 $^+$ phenotype (15). The authors reported expression of V γ 4 rather than V γ 6 in the majority (~50%) of cells, although markers for V γ 6 $^+$ cells were not assessed. The potential difference in V γ chain expression in CD18 KO mice versus CD18^{hy γ} mice could be due to genetic differences (C57BL/6J versus PL/J backgrounds), the level of CD18 expression (absent versus low), or changes induced by inflammation [no overt inflammation in our CD18 KO mice; severe dermatitis in the CD18^{hy γ} mice (15)].

Although thymic development of V γ 6 $^+$ cells was unaffected by β_2 integrin loss, we found an unexpected defect in V γ 4 $^+$ cell development. The defect was specific to the IFN γ -producing CD27 $^+$ V γ 4 $^+$ cells. The loss of CD18 impacts early in V γ 4 $^+$ $\gamma\delta$ T cell development (fewer immature V γ 4 $^+$ cells at E18.5) but is not absolutely necessary, as the V γ 4 $^+$ cells are reduced in number but not completely ablated. Interestingly, the numbers of V γ 4-expressing $\gamma\delta$ T cells were only reduced in the peripheral LNs of adult CD18 KO mice and not in the spleen or lungs where this population also resides. This may indicate that migration of V γ 4 $^+$ cells to peripheral LNs is partially dependent on β_2 integrins or that factors present in the spleen and lungs but absent in peripheral LNs preferentially expand or retain V γ 4 $^+$ cells. The functional consequences of the reduced numbers of V γ 4 $^+$ cells in peripheral LNs of CD18 KO mice remain to be determined. As IFN γ production by V γ 4 $^+$ cells is protective in models of melanoma (36, 37) and viral infection (38), reduced numbers of V γ 4 $^+$ cells may enhance pathology in these models.

Our study highlights survival, rather than proliferation, as the dominant mechanism of V γ 6 $^+$ cell accumulation in adult CD18 KO mice. We did find a slight but significant increase in Ki67 expression in V γ 6 $^+$ cells from CD18 KO mice, potentially indicating a longer time spent in active cell cycle (39) while not progressing to proliferation. Indeed, by BrdU incorporation we

found little evidence of active $\gamma\delta$ T cell proliferation regardless of genotype, which fits with published reports showing low turnover of V γ 6 $^+$ cells in adult mice (34, 40). Although it is possible that CD18 KO cells may be undergoing increased proliferation compared to WT at a slow rate not detected, the scRNAseq analysis shows a very small population of dividing (Cdk6 $^+$) $\gamma\delta$ T cells, the frequency of which is unaltered in CD18 KO mice. While it remains possible that proliferation contributes to an increase in V γ 6 $^+$ cells in neonatal mice, our data illustrate that survival is the major contributor to V γ 6 $^+$ population expansion in β_2 -integrin-deficient mice from 4 wk of age and in adult mice.

scRNAseq analysis of lung $\gamma\delta$ T cells revealed a striking enrichment of genes associated with survival, specifically in the β_2 -integrin-deficient V γ 6 $^+$ $\gamma\delta$ T cells. Importantly, ex vivo analysis confirmed that V γ 6 $^+$ cells from the lungs of CD18 KO mice displayed decreased apoptosis compared to WT. Several TCR components as well as molecules downstream from the TCR-signaling pathway (Cd44, Icos, Lat, Nr4a1 [Nur77]) were increased in CD18 KO V γ 6 $^+$ cells, suggesting that TCR signaling may contribute to the prosurvival phenotype. Although ligands that drive the expansion and retention of V γ 5 $^+$ and V γ 7 $^+$ $\gamma\delta$ T cells have been characterized, namely Skint1 (41) and Btl1 (42), respectively, a ligand specific for V γ 6 $^+$ cells has not yet been identified. If such a ligand exists, an up-regulation in CD18 KO lungs and uterus may contribute to the expansion phenotype seen in this study. Furthermore, two members of the Bcl2a1 family (Bcl2a1a, Bcl2a1d), which have been reported to support the survival of skin-resident V γ 6 $^+$ cells (29), were also enhanced in our CD18 KO lung V γ 6 $^+$ cells. Therefore, it is likely that a combination of signaling pathways that promote cell survival contributes to the accumulation of V γ 6 $^+$ cells in β_2 -integrin-deficient mice.

One remaining question is, why is the increase in $\gamma\delta$ T cells restricted to the V γ 6-expressing subset? The β_2 integrins expressed by $\gamma\delta$ T cells may contribute directly to the induction of apoptosis (43). As we found LFA-1 expression to be highest in V γ 6 $^+$ cells in the lungs, this may explain the specificity of survival of the V γ 6 $^+$ subset in LFA-1-deficient mice.

Although not definitive from our study, a cell-intrinsic role for LFA-1 in suppressing $\gamma\delta$ T cell apoptosis is supported by the following evidence: 1) $\gamma\delta$ T cell expansion was dependent on the loss of LFA-1 but of no other β_2 integrin family members; 2) in our hands, $\gamma\delta$ T cells highly expressed LFA-1 but no other CD11 subunits; 3) the expansion of $\gamma\delta$ T cells was apparent in the tissues of CD18 KO mice from 1 d post birth when other lymphocyte populations are still developing (e.g., very few $\alpha\beta$ T cells were present); and 4) the increase in $\gamma\delta$ T cells was first observed in the lungs where stromal populations (negative for LFA-1) rather than leukocytes predominate. We, therefore, propose a model in which LFA-1, potentially in conjunction with TCR and/or death receptor signaling, contributes to the induction of apoptosis in V γ 6 $^+$ $\gamma\delta$ T cells.

The functional consequences of the expanded IL-17-producing $\gamma\delta$ T cell population in β_2 -integrin-deficient mice remain to be determined. IL-17-producing $\gamma\delta$ T cells are known to be required for effective pathogen clearance, particularly in the lungs and uterus (5–7, 44), and yet contribute to pathology in models of autoimmune disorders (45, 46) and cancer (47). We predict that the expanded V γ 6 $^+$ cell population will alter the outcome in these immune challenge models in an IL-17-dependent manner.

Although a subset equivalent to the V γ 6 $^+$ $\gamma\delta$ T cells identified in mice does not exist in humans, future analysis of V δ 1- and V δ 2-expressing subsets, both of which have been shown to produce IL-17, may reveal a potential role for β_2 integrins in regulating specific subsets of human $\gamma\delta$ T cells. The targeting of IL-17-producing $\gamma\delta$ T cells has been identified as a therapeutic strategy for patients with spondyloarthritis (48); therefore further work to elucidate the role of β_2 integrins in the regulation human $\gamma\delta$ T cells may reveal novel targets for therapy.

In summary, our findings reveal unanticipated roles for the β_2 integrin LFA-1 in regulating the balance between IL-17[–] and IFN γ -producing $\gamma\delta$ T cell subsets: β_2 integrins promote the development of IFN γ -producing CD27⁺ V γ 4⁺ cells in the thymus, while suppressing the survival of IL-17[–]-producing CD27[–] V γ 6⁺ cells in the periphery. The striking expansion of V γ 6⁺ cells in β_2 -integrin-deficient mice demonstrates the central role that this integrin plays in regulating this elusive $\gamma\delta$ T cell population. The impact of these and other adhesion molecules on the survival and phenotype of $\gamma\delta$ T cells should be carefully considered when developing cancer therapeutics to target the immuno-protective functions of this lymphocyte population.

Materials and Methods

Mice. CD18 (#003329) KO mice were purchased from The Jackson Laboratory. Age- and sex-matched mice were used in all experiments. C57BL/6J mice (Envigo) or transgenic negative littermates were used as controls. Mice were housed at the University of Glasgow. All protocols were conducted under licenses issued by the UK Home Office under the Animals (Scientific Procedures) Act of 1986 and approved by the local ethics committee.

Tissue Processing. Single-cell suspensions were prepared from spleen, thymus, mesenteric LNs, mediastinal LNs, peripheral LNs (inguinal, brachial, and axillary pooled) and Peyer's patches by mashing between 45- μ m NITEX pieces (Cadisch Precision Meshes). Cells were isolated from bone marrow by flushing using a 26G needle. Spleen, blood, and bone marrow cell suspensions were lysed with red blood cell lysis buffer (eBioscience). Uteri were digested using Multi Dissociation Kit 1 (Miltenyi Biotec) according to the manufacturer's instructions. Ear skin was digested in 2 mg/mL Collagenase IV, 2 mg/mL hyaluronidase (both Sigma-Aldrich) and 100 U/mL DNase I (Invitrogen) at 37 °C for 40 min at 180 RPM in a rotating incubator and then dissociated using a gentleMACS (Miltenyi Biotec). The following tissues were digested as described: lungs (49), small intestine (50), and gingiva (51). Cells were counted by Trypan blue exclusion.

Flow Cytometry. Details of flow cytometry antibodies can be found in *SI Appendix*. Cells were stained with fixable viability dye eFluor506 (eBioscience) for 30 min at 4 °C. Extracellular staining was performed in Fc receptor blocking solution (supernatant from 2.4G2 cell line) for 30 min at 4 °C. For intracellular cytokine staining, cells were stimulated with 10 ng/mL phorbol myristate acetate (PMA; Sigma Aldrich), 500 ng/mL ionomycin (Sigma Aldrich), and 1:1,000 GolgiPlug (BD Biosciences) for 4 h at 37 °C. Following extracellular staining, cells were fixed using Cytofix/Cytoperm and stained in Perm/Wash (both BD Biosciences). For transcription factor staining, cells were fixed using FoxP3 Transcription Factor Fixation/Permeabilization buffer and stained in permeabilization buffer (both eBioscience). All intracellular staining was for 1 h at room temperature (RT). For apoptotic cell staining, cells were stained with AnnexinV (Biolegend) and 7AAD (BD Biosciences) in AnnexinV staining buffer (Biolegend) for 15 min at RT. All cells were resuspended in fluorescence-activated cell sorting (FACS) buffer (phosphate-buffered saline 2% fetal bovine serum (FBS) and 2 mM ethylenediaminetetraacetic acid) and acquired on LSRII or LSR Fortessa (BD Biosciences). Data were analyzed using FlowJo software (Treestar).

Fluorescence-Activated Cell Sorting. $\gamma\delta$ T cells from CD18 KO spleens were pre-enriched using the EasySep Mouse T cell Isolation Kit (Stemcell Technologies). Live CD3⁺TCR $\gamma\delta$ ⁺, V γ 1[–]V γ 4[–] cells were then sorted using a FACS Aria IIIU (BD Biosciences).

RNA Isolation, Complementary DNA, and PCR. RNA was purified using RNeasy Mini Kit (Qiagen) according to the manufacturer's instructions. Complementary DNA (cDNA) synthesis was carried out using High-Capacity RNA-to-cDNA Kit (Applied Biosystems). V γ and V δ chain sequences were

amplified using 0.02 U/ μ L, 1 $\times$ Phusion HF buffer, 3% dimethylsulfoxide, 2 mM MgCl₂ (V γ 6 reaction only) (all Thermo Fisher), 0.2 mM dNTPs (Invitrogen) and 0.3 μ M primers (IDT Technologies). Primer sequences for V γ 1, V γ 4, V γ 5, V γ 6, and V γ 7 were provided by Pierre Vantourout, King's College, London, and for V γ 2 and V δ have been described (52). DNA fragments were separated on 1% agarose (Invitrogen) gels and visualized using Gel Doc XR+ system (BioRad).

Single-Cell RNA Sequencing. $\gamma\delta$ T cells were isolated from the lungs of 2 WT mice and 1 CD18 KO mouse by FACS sorting for CD3⁺TCR $\gamma\delta$ ⁺ cells. Single-cell suspensions were loaded into a Chromium Controller (10 $\times$ Genomics). Subsequently, cDNA was generated from polyadenylated RNA, amplified, and converted to Illumina compatible libraries according to the standard protocol published in the Single Cell 3' Reagent Kits v3 User Guide (10 $\times$ Genomics). These were sequenced on an Illumina NextSeq500 sequencer with a sequencing depth of ~50,000 read pairs per cell. The read 1 and 2 were 28 and 91 bp long, respectively. scRNAseq reads were mapped in Cellranger (version 3.1.0) (53) against the current *Mus musculus* reference (mm10-3.0.0). The obtained readcount matrixes were analyzed in Seurat (54), version 3.1.2 in R version 3.5.3. We followed the analysis as described (55). To clean the data of doublets and dying cells, we filtered cells for a mitochondria amount less than 18% and applied the nFeature_RNA a cutoff of 2,200, 3,200, and 2,500 cells for WT1, WT2, and KO, respectively. To remove technical variation, we used the SCTransform protocol in Seurat. WT1 and KO samples were integrated first, and then WT2 was added which was down sampled to 4,000 cells. The number of PCA transformation dimensions was 15, and the "findcluster" resolution was 0.15. Marker genes were selected for cluster annotation, non-T cell clusters were removed, and clustering was then repeated. The default Seurat values (FindMarkers function) were used to determine DE genes, with an adjusted P value cutoff of 0.01. Venny 2.1 (<http://bioinfogp.cnb.csic.es/tools/venny/>) was used to compare DE genes in different clusters. GO enrichment pathway analysis was performed in Panther (pantherdb.org/). The reads and read counts are available in ArrayExpress under the accession number E-MTAB-8732. Furthermore, we designed a cell atlas to query the data, which can be accessed via this link: <http://cellatlas.mvls.gla.ac.uk/gdt-cell/>. This cell atlas was programmed in R-shiny using the Seurat pipeline as described above.

Statistical Analysis. Student's *t* test or two-way ANOVA (GraphPad Prism) were used to calculate statistical significance. Sidak's multiple comparisons test was used to correct for multiple comparison. *P* values shown are the following: ns, not significant; **P* < 0.05; ***P* < 0.01; ****P* < 0.005; and *****P* < 0.0001.

Data Availability Statement. The reads and read counts from the single-cell RNA sequencing are available at ArrayExpress (<https://www.ebi.ac.uk/arrayexpress/>) under accession number E-MTAB-8732. All other data are included in this article or *SI Appendix*. Further information about data and reagents used is available from the corresponding author.

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